THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. During a solar eclipse, small gaps in leaves form: 1

(a) full round images **(b)** crescent images of the Sun
(c) no images **(d)** square images

2. Which word looks exactly the same in a plane mirror (when the left and right are interchanged)? 1

(a) DAD **(b)** MAD **(c)** BOB **(d)** MOM

3. A large source of light produces: 1

(a) shadow with umbra and penumbra **(b)** only umbra
(c) bright shadow **(d)** no shadow

4. The size of a shadow does not depend on: 1

(a) distance from screen **(b)** size of the source
(c) the colour of the object **(d)** distance from source

5. A clock in a mirror reads 3:00. The mirror image looks like the real time of: 1

(a) 12:00 **(b)** 6:00 **(c)** 3:00 **(d)** 9:00

6. A sheet of tracing paper placed between source and screen: 1

(a) makes no light **(b)** makes dark shadow
(c) makes a faint shadow **(d)** makes a bright shadow

7. In which of the following will you see a shadow? 1

(a) a sunny field **(b)** a foggy night
(c) underwater at depth **(d)** a dark room

8. Which letter looks the same in a plane mirror? 1

(a) A **(b)** R **(c)** B **(d)** F

9. Which cannot cast a shadow? 1

(a) a perfectly transparent medium such as clean air
(b) a coin
(c) a book
(d) a ball

10. A kaleidoscope with two mirrors inclined at 60° forms ___ images of an object (basic). 1

(a) 6 **(b)** 2 **(c)** 5 **(d)** 1

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A clock shows 4:20 in a mirror. What is the actual time? 2

12. Why does the shadow of a flagpole change in length through the day? 2

13. Give two examples where diffused reflection is useful. 2

14. Why does a dress shop use mirrors on walls? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how the shadow of an object changes when it is placed between a bulb and a screen at different distances. 3

16. Explain the term 'real image' and say whether the image in a pinhole camera is real. 3

17. Explain why we see a rainbow-coloured sky glare on a wet road. 3

OR

Explain with a diagram in words how the periscope shows an object behind a wall. 3

18. Why does a book appear white when illuminated by white light but we see its text black? 3

OR

What is the nature of a shadow of a transparent glass plate? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss the use of mirrors at home, in vehicles, in instruments and in nature with examples. 5

OR

Explain what happens to the shadow of a person walking away from a street lamp. 5

20. Explain how umbra and penumbra form when a ball is placed between a large lamp and a screen. 5

OR

Describe how mirrors can be used in a solar cooker or solar torch. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND